

Code Coverage Exceptions & Justification

For this project phase, the JUnit testing suite focuses heavily on the core algorithmic logic and data filtering mechanics, successfully achieving high coverage on the AlertGenerator and DataStorage classes. However, a few specific components were intentionally excluded from the unit tests:

- **Main.java:** This class serves exclusively as the application's entry point. Its sole responsibility is to parse command-line arguments and route the execution to the appropriate system (DataStorage or HealthDataSimulator). Because it contains no standalone business logic and relies entirely on standard system inputs, it was excluded from unit testing.
- **FileDataReader.java (DataReader Implementation):** This component is responsible for direct I/O operations and file system parsing. Writing pure unit tests for this class would require relying on external file structures or heavy mocking of the java.io library, which falls outside the scope of testing the internal memory storage and alert evaluation logic.
- **Data Models (Alert.java, PatientRecord.java):** These are simple Plain Old Java Objects (POJOs) consisting only of fields, constructors, and getter methods. Since they do not contain complex behaviors or algorithmic logic, they do not require dedicated unit tests, though they are implicitly tested during the execution of the storage and generator tests.